Compositions of Substances & Solutions

pure elements
compounds

Homogenous mixture.

Chapter 6

Chemistry: Atoms First 2nd Ed.

Louis Pascal Riel
UConn Avery Point

Spring 2026

1127 Housekeeping Mar 17th Quiz 5 Topics

- HW see due dates →
  - Skipping A8, might circle back semester's end.

Due 3 Tuesdays!!

A9: Chapter 6 Compositions of Substances & Solutions
Homework
Start Date: Tuesday, Jan 13, 2026 7:20 PM
Due Date: Tuesday, Mar 31, 2026 11:59 PM
0 problems

DUE IN 21 DAYS
A10: Chapter 7 FIRST PORTION, Balancing Equations, Molecular Equations, Net Ionic Equations
Homework
Start Date: Tuesday, Jan 13, 2026 7:20 PM
Due Date: Tuesday, Apr 7, 2026 11:59 PM
30 problems

- 1-sided page of paper notes for a potential EC pop quiz Tuesday after break!

- Extra Credit – Not too early to start considering your group for Exam III Review Game
  - Reminder: at least 1 question must come from today's material!

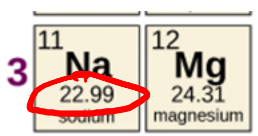
Chapter topics

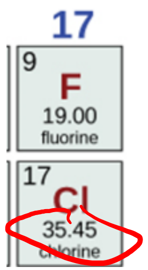
- A review:
- Formula Mass (~ **Molar Mass (MM) = Molecular Weight (MW)**)
- Empirical and Molecular Formulas

- Molarity
    - & other units for solution concentrations

- Dilution

Molar Mass (MM or MW)





- The **molar mass** is the sum of the average atomic masses of all the atoms in a substance's chemical formula

Fill in blanks & write conversion factor

- NaCl's molar mass is 22.99 + 35.45 = 58.44 (g NaCl / mol NaCl)

$$\frac{58.44\text{ g NaCl}}{1\text{ mol NaCl}}$$

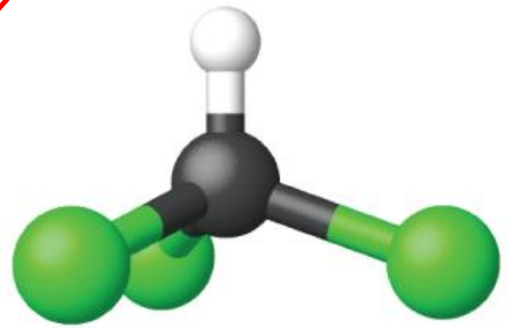
For this class, we will use **molar mass (MM)**
to mean molecular mass/molecular weight/formula mass/formula weight

Covalent Compound's Formula Masses

- The chemical formula for covalent compounds represents the amount & type of atoms composing a single molecule of the substance
  - ∴ Formula mass = molecular mass
- Chloroform, $CHCl_3$, a surgical anesthetic, and now used as a precursor to Teflon

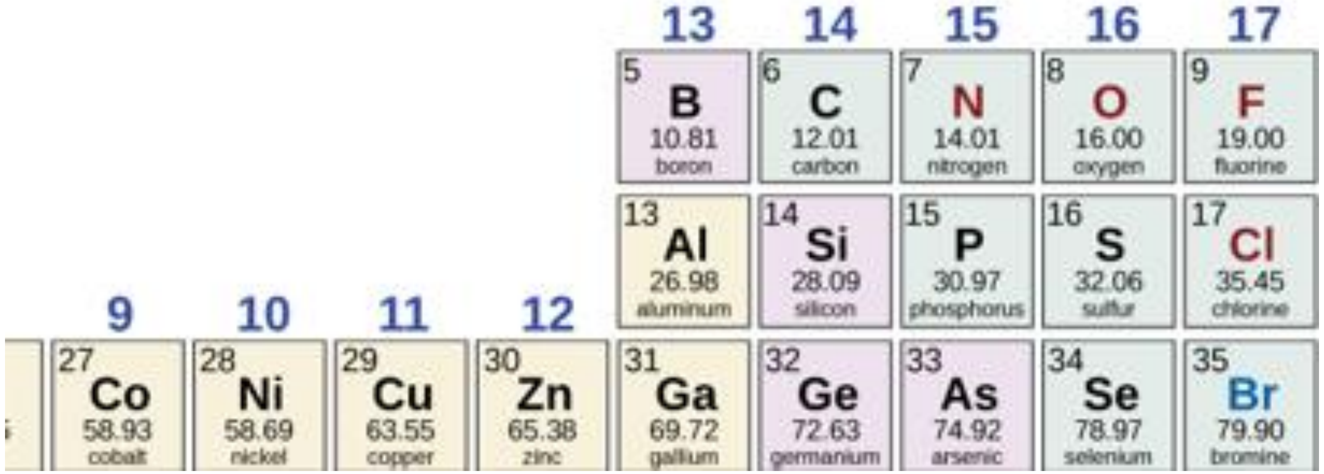
$$\frac{119.37\,g\ CHCl_3}{1\ mol\ CHCl_3}$$

Element	Quantity		Average atomic mass (amu)		Subtotal (amu)
C	1	×	12.01	=	12.01
H	1	×	1.008	=	1.008
Cl	3	×	35.45	=	106.35
			Molecular mass		119.37



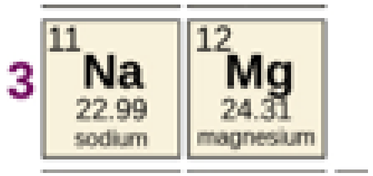
ICE MM

- 30 s



13
14
15
16
17
5 B 10.81 boron
6 C 12.01 carbon
7 N 14.01 nitrogen
8 O 16.00 oxygen
9 F 19.00 fluorine
13 Al 26.98 aluminum
14 Si 28.09 silicon
15 P 30.97 phosphorus
16 S 32.06 sulfur
17 Cl 35.45 chlorine
9
10
11
12
27 Co 58.93 cobalt
28 Ni 58.69 nickel
29 Cu 63.55 copper
30 Zn 65.38 zinc
31 Ga 69.72 gallium
32 Ge 72.63 germanium
33 As 74.92 arsenic
34 Se 78.97 selenium
35 Br 79.90 bromine

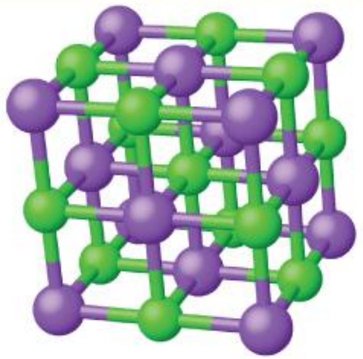
Molar Mass for Ionic Compounds



- Ionic compounds are composed of discrete cations and anions combined in ratios to yield electrically neutral bulk matter.

- There isn't a discrete molecule, so technically the term molar mass is incorrect, but we're going to use it anyway!

Element	Quantity		Average atomic mass (amu)		Subtotal
Na	1	×	22.99	=	22.99
Cl	1	×	35.45	=	35.45
Molar Mass ~ Formula mass					58.44



Recall/Practice: Determine Molar Mass of $Al_2(SO_4)_3$

Aluminum Sulfate

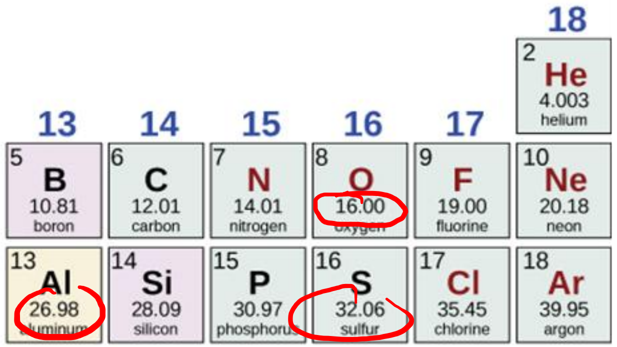
Answer on next slide

Recall/Practice: Determine Molar Mass of $Al_2(SO_4)_3$

S O$_4$

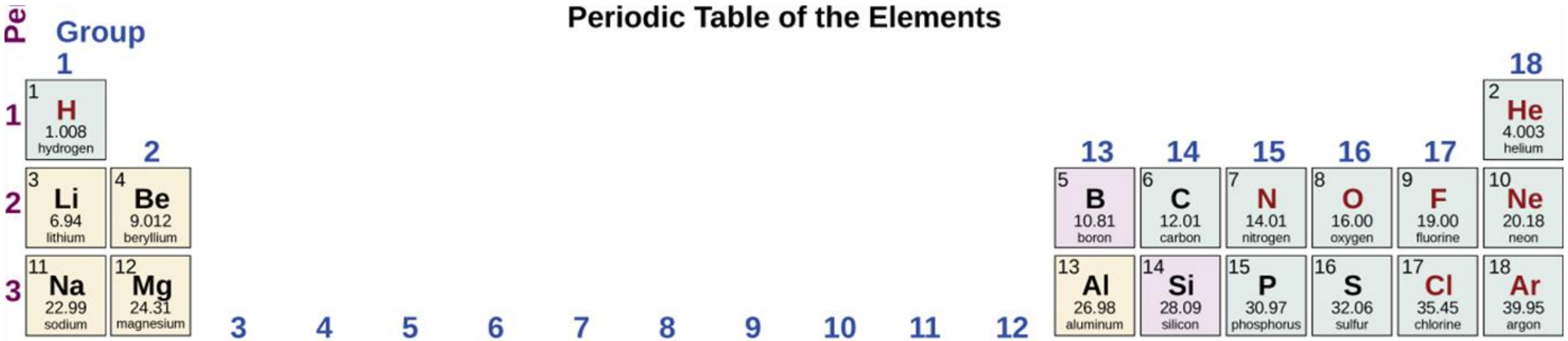
MM = 2 $\underbrace{(27)}_{Al_2}$ + 3 { 1 (32) + 4 (16) }

MM = 342 g $Al_2(SO_4)_3$ / 1 mol $Al_2(SO_4)_3$



ICE MM Ionic Compound

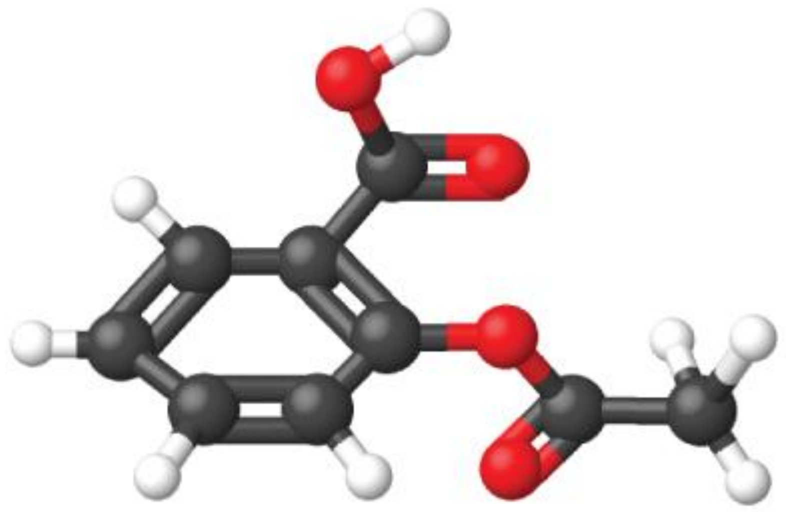
- 60 s



Aspirin’s Molar Mass

- Aspirin, $C_9H_8O_4$
- Molar Mass = 180.15 g Aspirin/ 1 mol Aspirin

Element	Quantity		Average atomic mass (amu)		Subtotal (amu)
C	9	×	12.01	=	108.09
H	8	×	1.008	=	8.064
O	4	×	16.00	=	64.00
			Molecular mass		180.15



When using <u>**molar masses**</u> in dimensional analysis problems, always <u>**1 mol**</u> in the fractions for MM or N_A

For H_2O: $\frac{18\ \mathrm{g}}{1\ \mathrm{mol}}$ or $\frac{1\ \mathrm{mol}}{18\ \mathrm{g}}$ or $\frac{1\ \mathrm{mol\ \ water}}{6.02 * 10^{23}\ \mathrm{molecules\ water}}$

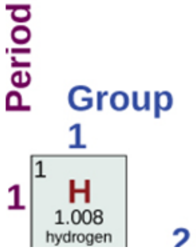
MM **N_A**

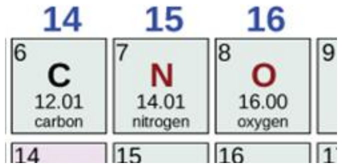
How many atoms of H are in 20 g H_2O?

$$\frac{20\,\cancel{g\,H_2O}}{1} \cdot \frac{1\,\cancel{mol\,H_2O}}{18\,\cancel{g\,H_2O}} \cdot \frac{2\,\cancel{mol\,H}}{1\,\cancel{mol\,H_2O}} \cdot \frac{6.02 \cdot 10^{23}\ \text{atoms H}}{1\,\cancel{mol\,H}} = 1.3 \cdot 10^{24}\ \text{atoms H}$$

MM ratio of elements N_a

ICE Use Molar Mass





- 90 s

- Requires you to calculate MM of $C_6H_{12}O_6$

- $N_A = 6.02 * 10^{23}$

100%

Percent Composition & Mass Percent

- Chemical formulas are **intensive properties** determined **empirically (via experimentation)**
  ↳ Does not depend on samplesize
- One way to Determine a formula is to take a sample of a pure compound and break it apart then separate it into its elements
  - $\text{Mass}_{\text{compound}}$ & the mass of the constituent elements are measured.
    - Knowing these two values allows us to determine the **mass percent** for each element in the compound.
      - **mass percent of ~~part~~ H** = $\frac{\textbf{mass of } \cancel{\textbf{part}}\ \text{H}}{\textbf{mass of total}} * \mathbf{100\%}$
        - For example, if a 10 g sample of a compound is 2.5 g H & 7.5 g C the mass percents would be

$$\%\,\mathrm{H} = \frac{2.5\,\mathrm{g\,H}}{10.0\,\mathrm{g\ compound}} \times 100\,\% = 25\,\%$$ H by mass

$$\%\,\mathrm{C} = \frac{7.5\,\mathrm{g\,C}}{10.0\,\mathrm{g\ compound}} \times 100\,\% = 75\,\%$$

Making Mass Percents into Conversion Factors

- For a given mass percent such as XX% part by mass in a mixture or compound

- Convert it to a ratio with: $\frac{XX\ g\ part}{100\ g\ YYY}$

  - Where YYY is the mixture or compound

- Denarius: initially 5% Cu by mass $\frac{5g\ Cu}{100g\ Coin}$
  - The rest silver, but then debasement

- Average human: 60 % H_2O by mass $\frac{60g\ H_2O}{100g\ Human}$

Sis felicior Augusto, melior Traiano

'be more fortunate than Augustus [and] better than __________'

Anhydrous means Without H2O

This will help you on a question that I *may* put on the quiz or exam from Aktiv Chemistry

Consider [$CuSO_4 \cdot 4\ H_2O$]

this means one unit of this compound has 1 $CuSO_4$ & 4 H_2O

MM of [$CuSO_4 \cdot 4\ H_2O$] equals

$$63.6 + 32.1 + 4\ (16) + 4\ (\ 2 * 1.0 + 16.0) = 231.67\ \text{g / mol}$$

Cu SO4 4 H2 O

4 H_2O

MM of **anhydrous** $CuSO_4$ = 159.6 g /mol

% mass H_2O = $\frac{231.67 - 159.6}{231.67} \cdot 100\%$

31.1% is H_2O

What are the 3 most useful numbers?

1 | 0 | 100

Unity | **The Abyss** | **24 % APR On one of my credit cards**

For H_2O: 1 **g /mL (density)** | 0 **°C (Freezing Point)** | 100 **°C (Boiling Point)**

Percent Composition (% X)

- For a compound/molecule,

$$\% X = \frac{\text{mass of element X in 1 mol of compound}}{\text{mass of 1 mol of compound}} * 100\%$$

  - Will never ask you for this definition

- To determine the percent composition, we must determine the mass percents of each element present.

Mass relations in chemical formulas

Problem type: What is the mass or percent composition for a compound?

requires formula of a compound → mass % of elements $_{\text{in compound}}$

method: For element X:

$$\% \underline{O} = \frac{\text{mass of } \underline{O_x} \text{ in 1 mol of compound}}{\text{mass of 1 mol of compound}} * 100\%$$

Fill in the blank

Key is assuming there is __1___ mol of compound present.

H_2O MM 18 g/mol

What is the percent composition of each element in H_2O?

$1 \text{ mol } H_2O \cdot \frac{18 g H_2O}{1 \text{ mol}} = 18 g H_2O$

$\frac{1 \text{ mol } H_2O}{1} \cdot \frac{1 \text{ mol } O}{1 \text{ mol } H_2O} \cdot \frac{16 g O}{1 \text{ mol } O} = 16 g O$

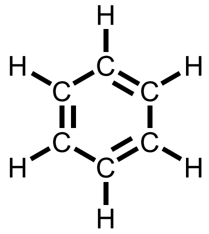
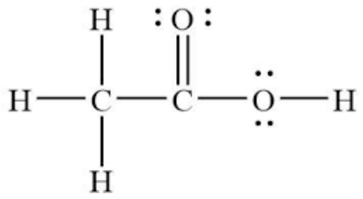
$\% O_x = \frac{16}{18} \cdot 100\% = 89\%$

$100\% - 89\% = \boxed{11\% H}$

This should be a review

Simplest or Empirical Formula (EF)

- A compound's EF is the **integer** formula formed from the **number of moles of each element** with all common factors removed from the subscripts for each element.
  - The simplest formula is what is first determined when finding formula

		Molecular Formula	EF
For Benzene:	 →	C_6H_6	CH
For Acetylene:	 →	C_2H_2	CH
For	 →	$C_2O_2H_4$	COH_2

Different Compounds *may* have the same EF & Percent Composition

- 10 g of pure benzene or pure acetylene each are composed of

- 10 g C_6H_6 $\dfrac{1\ \mathrm{mol}\ C_6H_6}{(6*12+6*1)\ g\ C_6H_6} * \dfrac{6\ \mathrm{mol}\ C}{1\ mol\ C_6H_6}\ \dfrac{12\ \mathrm{g}\ C}{1\ \mathrm{mol}\ C} =$ 9.2 g C

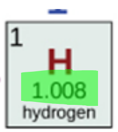
MM

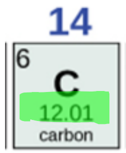
Compound's Elemental Ratio

- 10 g C_2H_2 $\dfrac{\underline{\ 1\ }\ \underline{\ mol\ }\ C_2H_2}{(2*12+2*1)\ \underline{\ g\ }\ C_2H_2} * \dfrac{\underline{\ 2\ }\ \underline{\ mol\ }\ C}{\underline{\ 1\ }\ \underline{\ mol\ }\ C_2H_2}\ \dfrac{\_\_\_\ \_\_\_ C}{\_\_\_\ \_\_\_ C} =$ 9.2 g C

Fill in blanks

Determination of Empirical Formulas





Consider an unknown compound that was measured to be 1.70 g C & 0.150 g H, **determine its empirical formula**.

simplest

Beginning w/ elements' masses, the steps are:

1. Convert mass to moles using average atomic mass for each element
   - Why? **Atoms/moles of each element** are combined in **integer mole ratios** to form compounds not g of each element.
2. These mole amounts become formula subscripts indicating *relative amounts*
3. Divide each number by smallest subscript results in empirical formula*

0.15g H $\div 1.008$ → 0.149 mol H

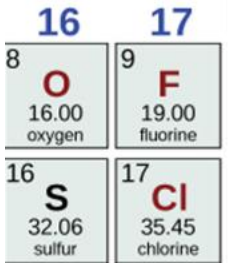
1.7g C $\div 12.01$ → 0.142 mol C

$H_{\frac{0.149}{0.142}} C_{\frac{0.142}{0.142}}$

$H_{1.05} C_1 \approx H_1 C_1$

integer mole ratio

Step 3* might result in fractional amount



Determine the empirical formula for a compound containing 5.30 g Cl & 8.40 g O.

$5.3 g Cl$ $8.4 g O$
$\div 35.45$ $\div 16$

1. Convert mass to moles using average atomic mass for each element

$0.1495 mol Cl$

$0.525 mol O$

2. These mole amounts become formula subscripts indicating *relative amounts*

$\frac{Cl_{0.1495}}{0.1495}$ $\frac{O_{0.525}}{0.1495}$

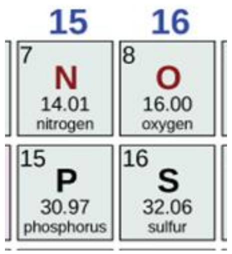
3. Divide each subscript by smallest subscript to provide empirical formula*
   - If the round seems too far, it likely is, instead, multiply each resulting subscript by an integer to make them all integers.

$Cl_1 O_{3.51}$
×2 ×2

Cl_2O_7

ICE Determine EF

- 2 mins



Spring 2026 – confirm what are applicable All!

Recorded Lecture Topics

- Calculate molar mass (MM)
- Determine percent compositions (mass percents)
- Identify simplest empirical formula
- Determine empirical formula from masses of each element

Empirical Formulas have **no** common multiples n or any decimals or fractions

CH_2 ~~not C_2H_4~~

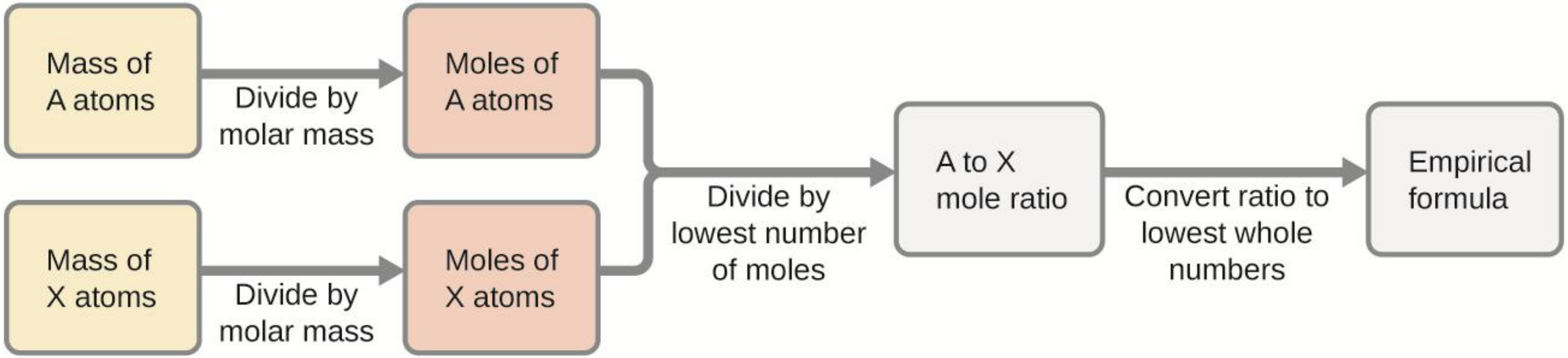
NO_2 ~~not N_2O_4~~

PO_5 ~~not $P_{0.5}O_{2.5}$~~

ICE EF Given MF

- 30 s

Process to Determine Empirical Formula



Microscopic Domain

Empirical Chemical Formulas tells the ratio of atoms of each element in a compound

Macroscopic Domain

Empirical Chemical Formulas tells the ratio of moles of each element in a mole of compound

Practice– worked out on next slide

MMs, Fe = 55.8 g/1 mol, O = 16 g/1 mol

A sample of the black mineral hematite, an oxide of iron found in many iron ores, contains 35.0 g of iron and 15.0 g of oxygen. What is the empirical formula of hematite?

worked out on next slide

Exam II Review - What is the proper IUPAC name for Hematite?

Practice

Answer w/ Work

MMs, Fe = 55.8 g/1 mol, O = 16 g/1 mol

A sample of the black mineral hematite, an oxide of iron found in many iron ores, contains 35.0 g of iron and 15.0 g of oxygen. What is the empirical formula of hematite?

Convert grams of iron & oxygen to moles

35.0 g Fe / 55.8 g/mol = 0.627 mol Fe, 15.0 g O / 16 g/mol = 0.938 mol O

Make these values stoichiometric subscripts → $Fe_{0.627}O_{0.938}$

Divide each by smaller value $Fe_{0.627/0.627}O_{0.938/0.627}$ = $Fe_1O_{1.5}$

Multiply by factor of 2 to clear decimals → Fe_2O_3

What is the proper IUPAC name for Hematite? Iron (III) Oxide

Mar 17
Quiz 5 topics

1127 Housekeeping Mar 24th Quiz 5 Topics

- HW see due dates →
  - Skipping A8, might circle back semester's end.

Due next Tuesdays

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Homework
Start Date: Tuesday, Jan 13, 2026 7:20 PM
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80 problems

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30 problems

- No pop quiz!

- Extra Credit – Not too early to start considering your group for Exam III Review Game
  - Reminder: at least 1 question must come from today's material!

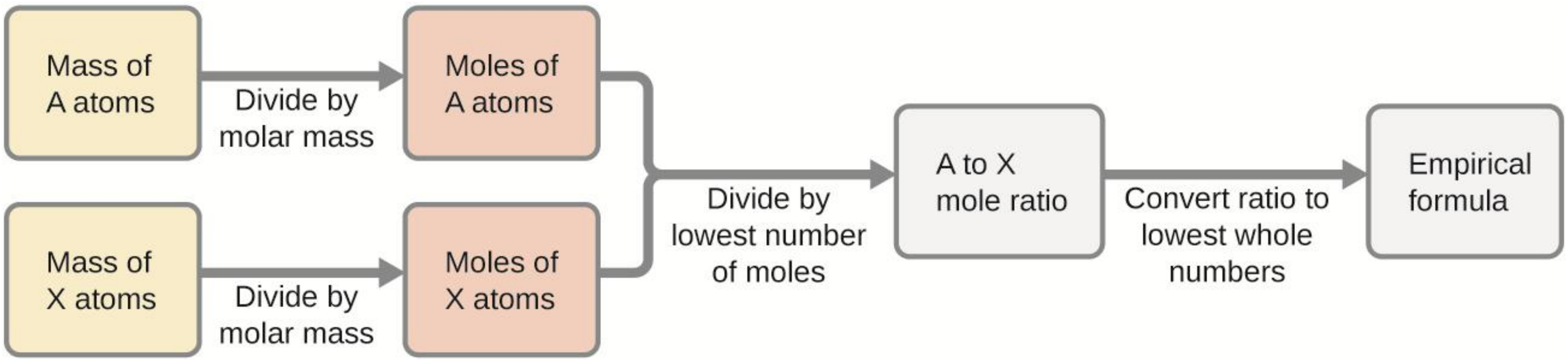
- Quiz 5 – Next Thurs Apr 2nd

Empirical Formulas from Percent Composition

- The easiest number to begin percent problems is _____
  - '_______%'
    - For converting percent composition to empirical formula:
      - assume that you have 100 g of the compound, this has the effect of converting % to g
- The bacterial fermentation of grain to produce ethanol forms a gas with a percent composition of 27.3% C and 72.7% O. What is the empirical formula for this gas?

Process to Determine Empirical Formula from *Percent Composition/Masses*

- *adds one step* from mass percents
- if 30% A and 70% X by mass



Empirical Formula | Molar Mass

Determining Molecular Formula from EF & MM

- To determine a molecular formula, a compound's MM and EF must be known/provided

- A compound has an empirical formula of CH_2 and its molar mass was determined to be 84 g/mol, determine its molecular formula.
  - Key step: divide given MM by empirical formula's molar mass to get a factor.

- Multiplying each subscript of the formula by this factor → MF

Built in Check – recalculate the MM of the resulting MF

ICE Determine MF from EF & MM

- 60 s, use previous slide!

Practice

Molar Mass = g substance /moles of substance

- **Determination of the Molecular Formula for Nicotine**

Nicotine, an alkaloid in the nightshade family of plants that is mainly responsible for the addictive nature of cigarettes, contains 74% C, 8.7% H, and 17.3% N. If 40.6 g of nicotine contains 0.25 mol nicotine, what is the molecular formula?

TRY THIS

Answer in Notes

Substances vs. **Mixtures**.

- Recall: a (pure) **substance** contains only a single type of element or compound
  - Cl_2, He, H_2O, Al_2O_3, $C_6H_{12}O_6$, NaCl

- **Mixtures** are samples containing two or more substances that have been **physical combined**.
- **Solutions** are **homogeneous mixtures** that have the same composition (& thus the same properties) throughout.

Solutions (Homogeneous Mixtures)

(aq) Salt Water
(Brine)

- **Solutions** (abbreviated as **soln**) are homogenous mixtures and are composed of a:

H_2O

- **Solvent** – the species that is most present/abundant.
  - The 'medium' into which the other species are **dissolved**.
  - **Aqueous <u>(aq)</u>** solutions are generally those where **water is the solvent**.

NaCl
Salt
Sodium Chloride

- **Solute(s)** – the specie(s) present in lower amounts.
  - there may be multiple solutes.

70% Ethanol (v/v) for Bactericidal Effects

- **Cidal**™ is a solution of 70% **ethanol** (volume by volume, v/v) in **water**. The company claims this is the optimal concentration of ethanol to kill bacteria. If 100 mL of solution are present, identify the **solute** and **solvent** and **solution**.

- Solution

- Solvent

- Solute

Dilute vs. Concentrated

New London & Manila
Population Densities (concentrations)

4,868/sq mi 112,953/sq mi

- **Dilute** solutes are present at low or small concentrations.
  - Gold ions (Au^{3+}) in the oceans.
    - about one gram of gold for every 100 million metric tons – quiz question??
- **Concentrated** solutes are present at high or large concentrations.
  - Sodium (___) or chloride (___) ions in the oceans.
    - 10561 g of the cation in 1,000,000 g of solution (seawater) – quiz question??
      - These two ions make up over 90% of all dissolved ions in seawater.
- Units of concentration allow us to quantitatively assess how much of a solute is present.

Molarity (**M**) – **THE** unit of concentration

- Molarity (abbreviated as M) or molar concentration is a unit of concentration defined as the number of moles of a solute (or any species) in 1 liter of solution.
    - A 6 M glucose solution has 6 mol glucose/ ___liter of solution

The highlighted parts mean the same thing

- 6 mol glucose per liter of solution

$$M = \frac{\text{mol solute}}{\text{L solution}}$$

Internalize

$$\boxed{\text{Molarity (M)} = \frac{\text{moles}}{\text{L}}}$$

- Helpful to use a given M in the **fractional form**, as a factor in **dimensional analysis**.
- Question: consider a 6 M aqueous ethanol (EtOH) solution:

1. How many moles are in 0.5 L?
    - (L → mol)
  - Or

$$\frac{0.5\ \text{L}}{1}\ \frac{6\ \text{mol EtOH}}{1\ \text{L}} = \text{3 mol EtOH}$$

2. What volume contains 2.4 mol EtOH?
    - (mol → L)

$$\frac{2.4\ \text{mol EtOH}}{1}\ \frac{1\ \text{L}}{6\ \text{mol EtOH}} = \text{0.4 L soln}$$

Molarity problems are often given 2 values, solve for the 3rd

1. Given $\text{volume}_{\text{solution}}$ & $\text{moles}_{\text{solute}}$ (or $\text{mass}_{\text{solute}}$), solve for molarity.

2. Given $\text{moles}_{\text{solute}}$ and molarity, solve for $\text{volume}_{\text{solution}}$.

3. Given $\text{volume}_{\text{solution}}$ & molarity, solve for $\text{moles}_{\text{solute}}$ (or $\text{mass}_{\text{solute}}$).

- Volumes may be in L or mL!
  - 10^{-3} L = 1 mL or 1000 mL = 1 L

$$\text{Molarity (M)} = \frac{\text{moles}}{\text{L}}$$

L to mL - shortcuts

- 1000 mL = 1 L
    - Shift decimal place 3 left for mL → L
- 750 mL = _____ L
- 80 mL = _____ L
    - Shift decimal place 3 right for L → mL
- 2.64 L = _____ mL
- 0.64 L = _____ mL

1 L (32 oz) bottle

Given on Quizzes & Exams

(in place of Exam I table, until Final Exam)

1 kg = 1000 g   1000 mL = 1 L

Will not be given on Assessments

Fill in

$$M = \underline{\qquad\qquad}$$

For Molarity:
L not mL
Liters not milliliters

Practice

- A 355-mL soft drink sample contains 0.133 mol of sucrose (table sugar). What is the molar concentration (molarity) of sucrose in the beverage?

Will be given on Assessments

1 kg = 1000 g

1000 mL = 1 L

ICE Determine Molarity

- 45 s

$$\mathrm{M} = \frac{\text{moles}}{\text{L}}$$

1 kg = 1000 g   1000 mL = 1 L

Practice

- How much sucrose (‘suc’, MM = 342 g/mol) **<u>in grams</u>** is contained in a modest sip (~10 mL) of the previous soft drink, that was 0.375 M sucrose?

$$M = \frac{\text{mol solute}}{\text{L solution}}$$

1 kg = 1000 g

1000 mL = 1 L

Practice

MM acetic acid = 60.1 g/mol

You need to be able to get the above MM from the formula below

- Distilled white vinegar is a solution of acetic acid (**AA**), CH_3CO_2H, in water. A 0.5 L vinegar solution contains 25 g of acetic acid. What is the concentration of the acetic acid solution in units of molarity?

$$M = \frac{\text{mol solute}}{\text{L solution}}$$

1 kg = 1000 g

1000 mL = 1 L

Practice 6.12

MM acetic acid = 60.1 g/mol

- The concentration of acetic acid in white vinegar was determined to be 0.84 *M*. What volume of vinegar, in mL, contains 76 g of acetic acid?

1 kg = 1000 g

1000 mL = 1 L

$N_A = 6.02 * 10^{23}$

Chemical Formula: moles vs atoms/molecules

- How many moles of H are in 12 mol glucose ($C_6H_{12}O_6$)?
- How many Li$^+$ ions are in 125 mL of 0.825 M Li_3PO_4?
- How many H atoms are in 1 unit (~ 1 molecule) of $(NH_4)_2HPO_4$
- How many H atoms are in 7.5 * 10^{-3} mol $(NH_4)_2HPO_4$?

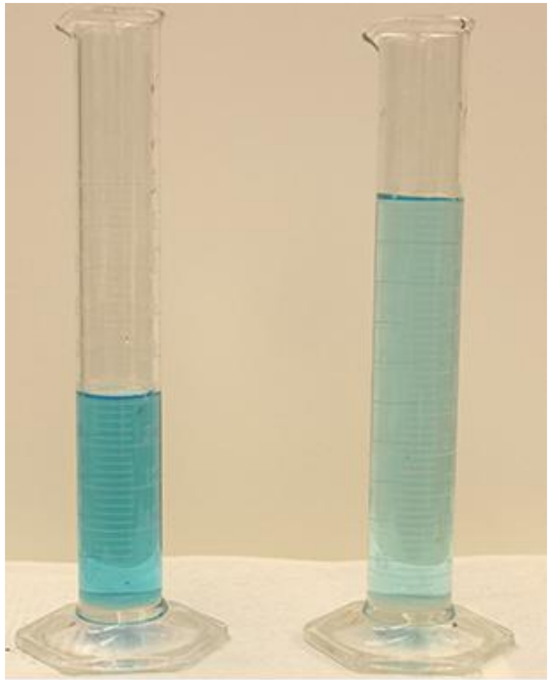
If all but one percents given,
the remainder is that last value

Key Types of *New* Problems – Chapter 6 So Far

- Percent Composition
  - Akin to “What is the percent composition of H_2O? “
- Empirical Formulas
  - Akin to “Consider an unknown compound that was measured to be 1.7 g C & 0.29 g H, **determine its empirical formula**.”
    - Or from mass percents!!
- Molecular formulas
  - Akin to “Nicotine is 74% C, 17.3 % N, and the remainder H by mass. If 40.6 g of nicotine contains 0.25 mol nicotine, what is the molecular formula?””
- Molarity Questions

Dilution of Solutions

- **Dilution** is the process whereby a solute's concentration decreases due to the addition of pure solvent.
  - If both graduated cylinders contain the same amount of $Cu(NO_3)_2$, which one is more concentrated **(MC)?**
- Dilution often starts from a *stock solution* of known concentration to which pure solvent is added to form less concentrated solutions.
- $C_i V_i = C_f V_f$ relates the premise that $\text{moles}_{\text{solute}}$ is fixed in the above case.

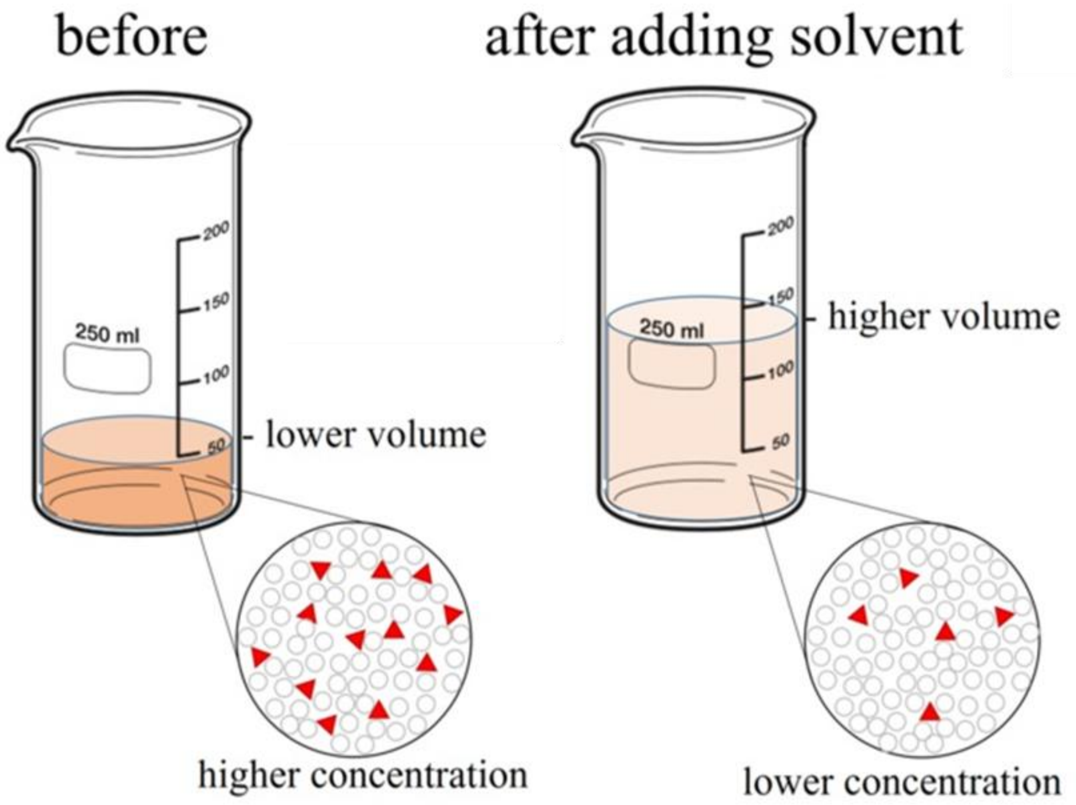


Stock Solution

Dilute Solution

$C_i V_i = C_f V_f$

How much water (pure solvent) was added?



- C_i = initial concentration (generally molarity)
- V_i = initial volume (can be L or mL)
- C_f = final concentration (generally molarity)
- V_f = final volume (can be L or mL)
  - V_1 & V_2 must be in the same units & C_1 & C_2 must be in the same units

You will not be given $C_i V_i = C_f V_f$
nor $M = mol / L$

$C_i V_i = C_f V_f$

If 0.85 L of a 5 *M* solution of copper nitrate, $Cu(NO_3)_2$, is diluted to a volume of 1.8 L by the addition of water, what is the molarity of the diluted solution?

Practice :

$$C_i V_i = C_f V_f$$

- What volume, in milliliters, of 0.12 M HBr can be prepared from 11 mL of 0.45 M HBr?

Practice:

$$C_i V_i = C_f V_f$$

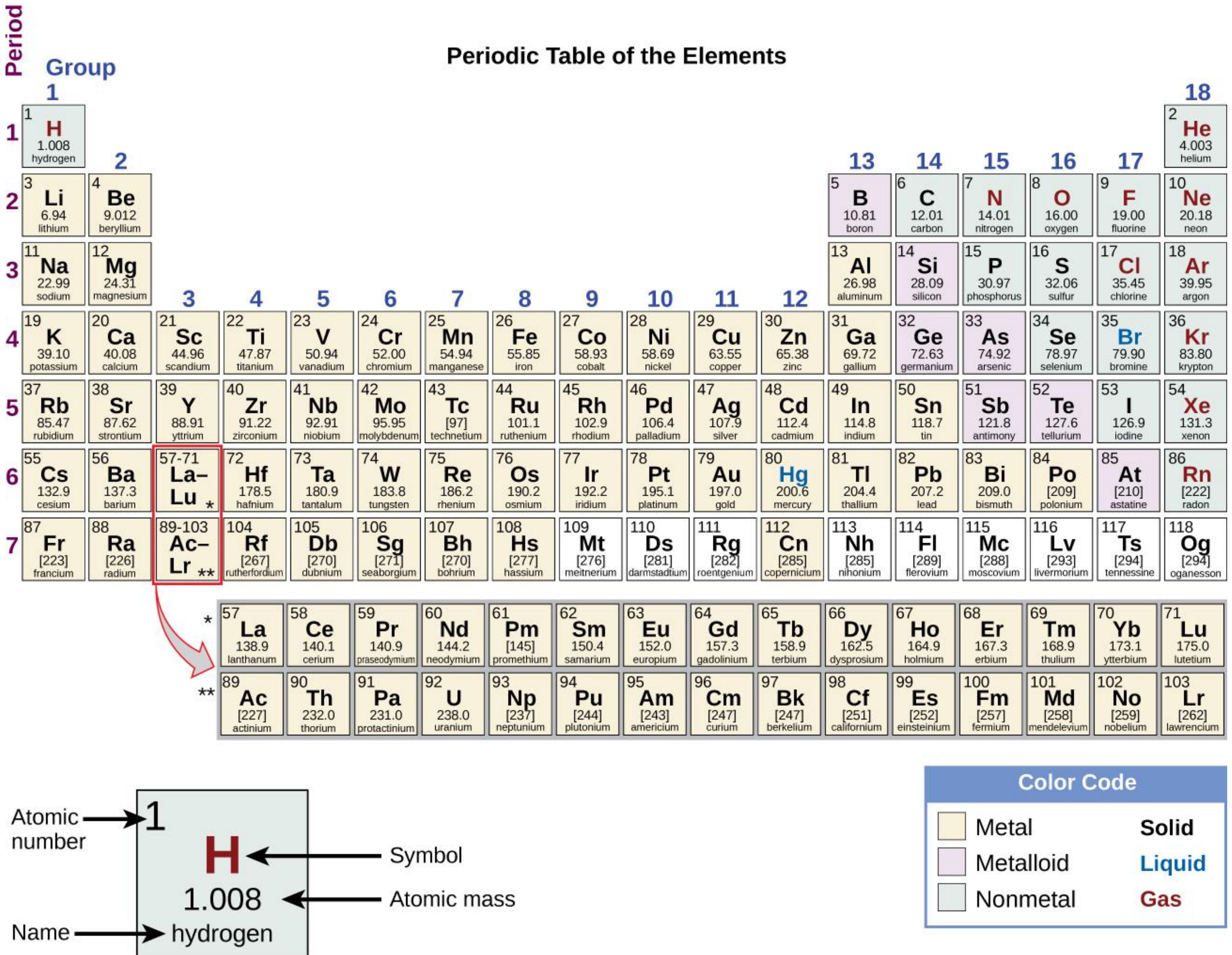
- What volume of 1.59 M KOH is required to prepare 5.00 L of 0.100 M KOH?

  - What volume of pure water must have been added to the stock solution?

Initial Final

ICE $C_i V_i = C_f V_f$

- 60 s


Periodic Table of the Elements
Period
Group
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
1 H 1.008 hydrogen
2 He 4.003 helium
3 Li 6.94 lithium
4 Be 9.012 beryllium
5 B 10.81 boron
6 C 12.01 carbon
7 N 14.01 nitrogen
8 O 16.00 oxygen
9 F 19.00 fluorine
10 Ne 20.18 neon
11 Na 22.99 sodium
12 Mg 24.31 magnesium
13 Al 26.98 aluminum
14 Si 28.09 silicon
15 P 30.97 phosphorus
16 S 32.06 sulfur
17 Cl 35.45 chlorine
18 Ar 39.95 argon
19 K 39.10 potassium
20 Ca 40.08 calcium
21 Sc 44.96 scandium
22 Ti 47.87 titanium
23 V 50.94 vanadium
24 Cr 52.00 chromium
25 Mn 54.94 manganese
26 Fe 55.85 iron
27 Co 58.93 cobalt
28 Ni 58.69 nickel
29 Cu 63.55 copper
30 Zn 65.38 zinc
31 Ga 69.72 gallium
32 Ge 72.63 germanium
33 As 74.92 arsenic
34 Se 78.97 selenium
35 Br 79.90 bromine
36 Kr 83.80 krypton
37 Rb 85.47 rubidium
38 Sr 87.62 strontium
39 Y 88.91 yttrium
40 Zr 91.22 zirconium
41 Nb 92.91 niobium
42 Mo 95.95 molybdenum
43 Tc [97] technetium
44 Ru 101.1 ruthenium
45 Rh 102.9 rhodium
46 Pd 106.4 palladium
47 Ag 107.9 silver
48 Cd 112.4 cadmium
49 In 114.8 indium
50 Sn 118.7 tin
51 Sb 121.8 antimony
52 Te 127.6 tellurium
53 I 126.9 iodine
54 Xe 131.3 xenon
55 Cs 132.9 cesium
56 Ba 137.3 barium
57-71 La–Lu *
72 Hf 178.5 hafnium
73 Ta 180.9 tantalum
74 W 183.8 tungsten
75 Re 186.2 rhenium
76 Os 190.2 osmium
77 Ir 192.2 iridium
78 Pt 195.1 platinum
79 Au 197.0 gold
80 Hg 200.6 mercury
81 Tl 204.4 thallium
82 Pb 207.2 lead
83 Bi 209.0 bismuth
84 Po [209] polonium
85 At [210] astatine
86 Rn [222] radon
87 Fr [223] francium
88 Ra [226] radium
89-103 Ac–Lr **
104 Rf [267] rutherfordium
105 Db [270] dubnium
106 Sg [271] seaborgium
107 Bh [270] bohrium
108 Hs [277] hassium
109 Mt [276] meitnerium
110 Ds [281] darmstadtium
111 Rg [282] roentgenium
112 Cn [285] copernicium
113 Nh [285] nihonium
114 Fl [289] flerovium
115 Mc [288] moscovium
116 Lv [293] livermorium
117 Ts [294] tennessine
118 Og [294] oganesson
*
57 La 138.9 lanthanum
58 Ce 140.1 cerium
59 Pr 140.9 praseodymium
60 Nd 144.2 neodymium
61 Pm [145] promethium
62 Sm 150.4 samarium
63 Eu 152.0 europium
64 Gd 157.3 gadolinium
65 Tb 158.9 terbium
66 Dy 162.5 dysprosium
67 Ho 164.9 holmium
68 Er 167.3 erbium
69 Tm 168.9 thulium
70 Yb 173.1 ytterbium
71 Lu 175.0 lutetium
**
89 Ac [227] actinium
90 Th 232.0 thorium
91 Pa 231.0 protactinium
92 U 238.0 uranium
93 Np [237] neptunium
94 Pu [244] plutonium
95 Am [243] americium
96 Cm [247] curium
97 Bk [247] berkelium
98 Cf [251] californium
99 Es [252] einsteinium
100 Fm [257] fermium
101 Md [258] mendelevium
102 No [259] nobelium
103 Lr [262] lawrencium
Atomic number
1
H
1.008
hydrogen
Symbol
Atomic mass
Name
Color Code
Metal
Metalloid
Nonmetal
Solid
Liquid
Gas